

Justin Manoj

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EDUCATION

University of Massachusetts - Amherst

Amherst, MA

Bachelor of Science in Computer Science & Mathematics | Honors College | GPA: 3.7/4.0

Aug. 2023 – May 2027

EXPERIENCE

Full-Stack Software Engineer

April 2025 – Present

ChitChat Workplace

Boston, MA

- Unified chat, email, video, and calendar into one workspace using **Next.js**, **React**, **TypeScript**, and **GraphQL API**
- Reduced context-switching by **10+ hrs/month** with edge microservices and real-time sync via **Kafka + WebSockets**
- Built **AI Copilot** with **GPT-4o**, **LangChain**, and **Pinecone** to power smart replies, summaries, and notifications
- Led deployment and monitoring of production microservices with **Kubernetes** and **Docker** to run **LiveKit + Redis Stream** pipeline handling **5K+ real-time messages/day**

Software Engineer Intern

March 2025 – Sept. 2025

GBCS Group/SkyIT Services

Alberta, CA

- Migrated **8 REST endpoints** to **GraphQL** to eliminate overfetching and shrink client requests on internal SaaS platform
- Optimized React Query, code splitting, and PostgreSQL indexes to cut page load times by **30%**
- Delivered **60+ reusable TypeScript/Tailwind components** and enforced code quality across 100+ PRs by implementing TypeScript linting rules, Jest utilities, and component documentation

Undergraduate Student Researcher

Dec. 2024 – Feb. 2025

Early Research Scholars Program - UMass Amherst

Amherst, MA

- Developed **Python framework** to detect malicious prompts in 500+ test cases, reducing manual review time by **70%**
- Enabled repeatable regression tests for **GPT-4**, **Claude**, and **Llama** with parameterized cases and standardized outputs
- Tested **10+ LLM attack vectors** and proposed security hardening recommendations for production deployment

PROJECTS

UMass Dining Recommendation Engine | Python, FastAPI, Next.js, PyTorch, Redis, PostgreSQL, Scikit-learn

- Built ML recommendation system to help students discover meals across 100+ menu items daily using **PyTorch** and **sentence-transformers** for personalized dietary-preference matching
- Deployed a FastAPI service handling **120-200 daily requests**; added Redis caching to cut latency by **~55%** (2.1s→0.9s)
- Implemented a custom embedding pipeline and ϵ -greedy multi-armed bandit for improving recommendation accuracy

Terminal Clipboard History Manager | Go, Bubble Tea, Lipgloss, Chroma, JSON

- Developed cross-platform (Linux, macOS, Windows) terminal clipboard manager with **Go** using **Bubble Tea TUI**, enabling users to browse, search, and manage past clipboard history
- Implemented JSON-backed storage, fuzzy search, auto-categorization, and syntax highlighting via Chroma, supporting up to **500-600 text/image entries** with duplicate detection and export functionality

Rust Utility Kit | Rust, Clap, QRCode, BLAKE3, Serde, Base64

- Built a **6-in-1 CLI** (URL shortener, password+QR generator, file organizer, base converter, text hasher) via **Rust + Clap**
- Implemented local URL shortening with SHA-256 hashing and JSON storage, QR code generation supporting PNG/ASCII output, and file organizer with dry-run mode supporting 7+ file categories and 40+ extensions

Collaborative Sticky Notes Application | TypeScript, React, Next.js, Liveblocks, Clerk, Docker, Prisma, PostgreSQL

- Built sticky notes application with **Next.js**, **TypeScript**, **React**, and **PostgreSQL** database integration, featuring **CRUD operations** for seamless note creation, editing, and deletion, supporting **up to 200 notes per user**
- Enabled real-time collaboration for **20-30 users** using **Liveblocks**, with **Clerk** auth and **Prisma + Docker** migrations

TECHNICAL SKILLS

Languages: Python, Java, Go, Rust, C/C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Tools: React, Next.js, Node.js, Express, FastAPI, Flask, Redux, Tailwind CSS, Jest, Postman

Cloud, Infrastructure & DevOps: AWS, Azure, Kubernetes, Docker, CI/CD, Kafka, Git/GitHub, OAuth 2.0

Data Systems, Machine Learning, & APIs: PostgreSQL, MongoDB, MySQL, Redis, GraphQL, Prisma, REST APIs; PyTorch, Scikit-learn, Pandas, NumPy, Tableau